Four-Slot Smart Parking Occupancy and Gate Controller ASIC

Static Timing Analysis (STA) - 18 Liberty File Runs

Objective: Evaluate setup timing, hold timing and total negative slack (TNS) using SKY130 HD timing libraries.

Design clock: 10 ns (100 MHz) **Clock uncertainty:** 0.2 ns **Input/output delay:** 1.0 ns

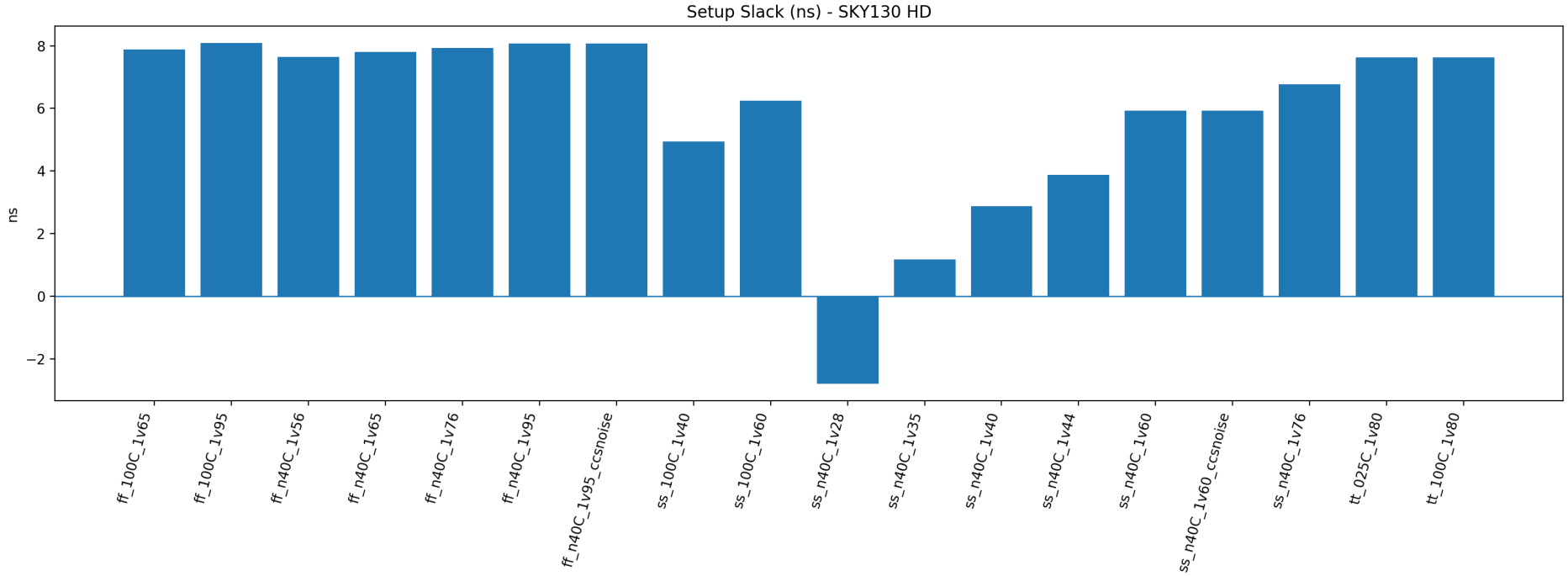
18-Run Timing Summary

Library Run	Setup Slack (ns)	Hold Slack (ns)	TNS (ns)
sky130_fd_sc_hd__ff_100C_1v65	7.89	0.28	0.0
sky130_fd_sc_hd__ff_100C_1v95	8.09	0.18	0.0
sky130_fd_sc_hd__ff_n40C_1v56	7.65	0.32	0.0
sky130_fd_sc_hd__ff_n40C_1v65	7.8	0.27	0.0
sky130_fd_sc_hd__ff_n40C_1v76	7.93	0.21	0.0
sky130_fd_sc_hd__ff_n40C_1v95	8.08	0.15	0.0
sky130_fd_sc_hd__ff_n40C_1v95_ccsnoise	8.08	0.15	0.0
sky130_fd_sc_hd__ss_100C_1v40	4.94	0.19	0.0
sky130_fd_sc_hd__ss_100C_1v60	6.25	0.41	0.0
sky130_fd_sc_hd__ss_n40C_1v28	-2.8	-0.9	-3.91
sky130_fd_sc_hd__ss_n40C_1v35	1.17	-0.34	0.0
sky130_fd_sc_hd__ss_n40C_1v40	2.88	-0.1	0.0
sky130_fd_sc_hd__ss_n40C_1v44	3.88	0.04	0.0
sky130_fd_sc_hd__ss_n40C_1v60	5.92	0.35	0.0
sky130_fd_sc_hd__ss_n40C_1v60_ccsnoise	5.92	0.35	0.0

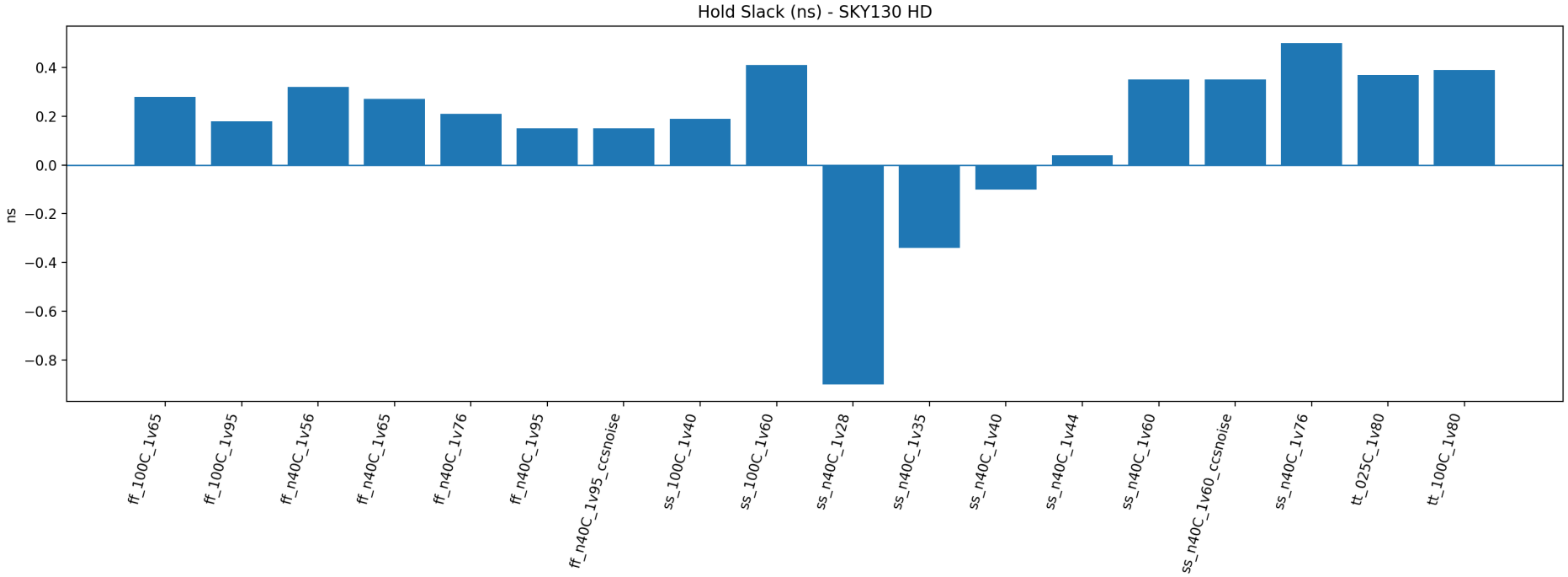
Library Run	Setup Slack (ns)	Hold Slack (ns)	TNS (ns)
sky130_fd_sc_hd__ss_n40C_1v76	6.77	0.5	0.0
sky130_fd_sc_hd__tt_025C_1v80	7.62	0.37	0.0
sky130_fd_sc_hd__tt_100C_1v80	7.63	0.39	0.0

Important result: The SS n40C 1v28 library run shows negative setup slack (-2.80 ns), negative hold slack (-0.90 ns) and TNS (-3.91 ns). This is the most critical timing case in the analyzed runs.

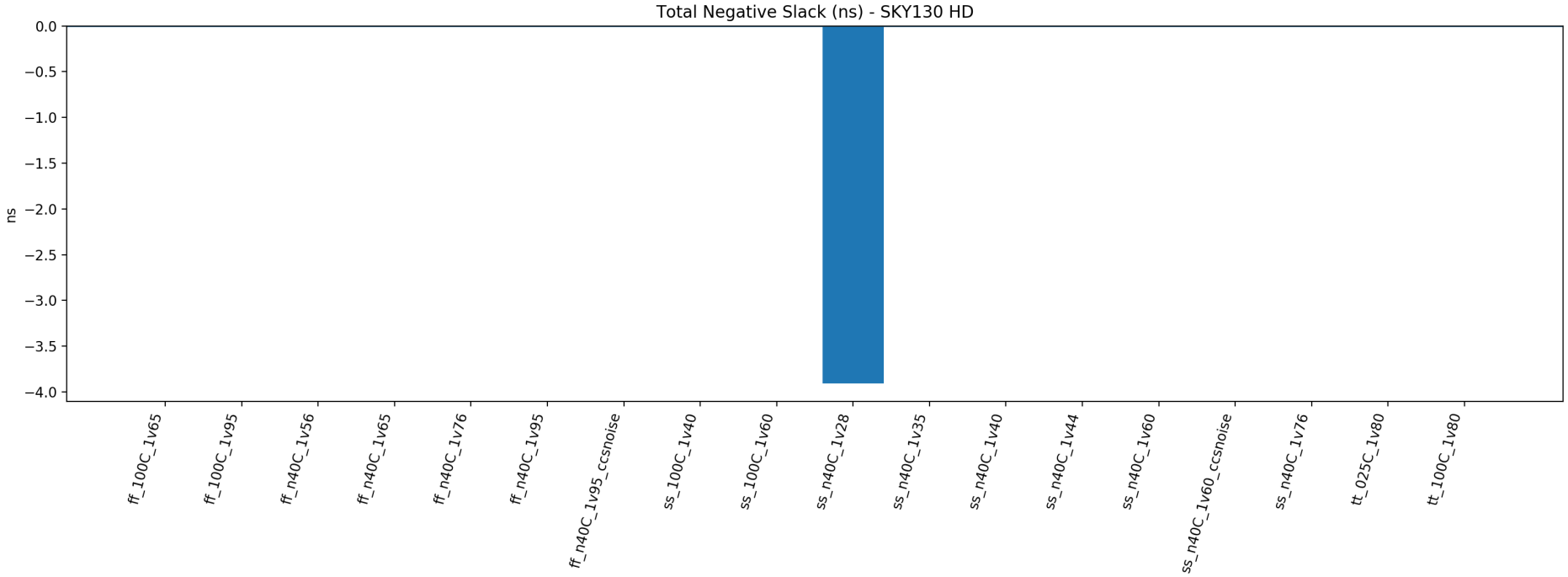
Setup Slack Across 18 Library Runs



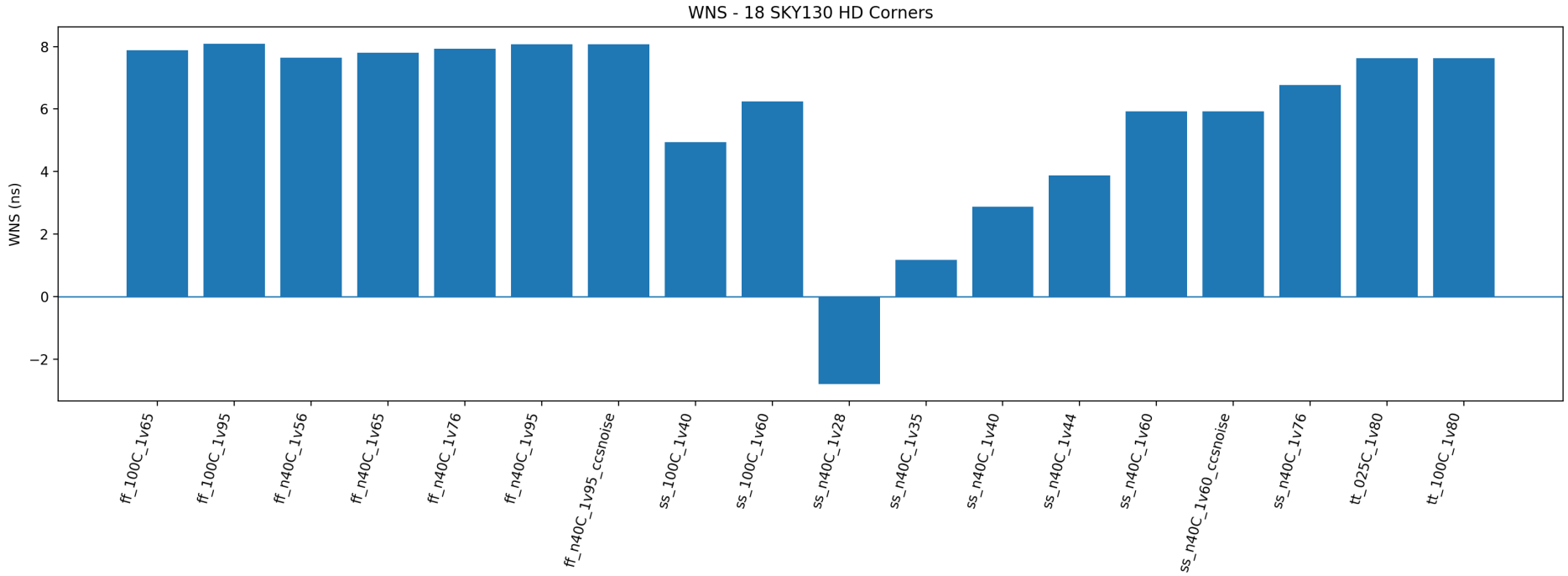
Hold Slack Across 18 Library Runs



Total Negative Slack (TNS) Across Runs



Worst Negative Slack (WNS) Across Runs



Conclusion and Limitations

The timing analysis was executed using OpenSTA and SKY130 HD Liberty files. Most runs show positive setup and hold slack, while selected low-voltage slow-process runs show timing violations. These results indicate that timing closure must be evaluated at the worst operating condition. This report is an educational analysis and is not foundry sign-off.

Note: The 18 runs represent 18 library files; CCS-noise variants may duplicate timing values and are not necessarily 18 unique PVT points.